ChatGPT

Graphs for Forecasting

Executive summary

Graph-based forecasting has evolved from **road-network-specific spatio-temporal models** into a broader family of methods for **general multivariate time series**, where each variable is treated as a node and the model either uses a known graph or learns one from data. The most influential early wave used graph convolutions plus recurrent or temporal-convolution modules, especially for traffic forecasting; key exemplars include DCRNN, STGCN, and Graph WaveNet. A second wave focused on **adaptive graph learning** and **node-specific modeling**, as in MTGNN and AGCRN. A third wave introduced **attention and spectral formulations** such as GMAN and StemGNN, while more recent work explores **continuous-time formulations** like STG-NCDE and CEGNCDE, along with scalable graph learners for larger sensor networks. [1]

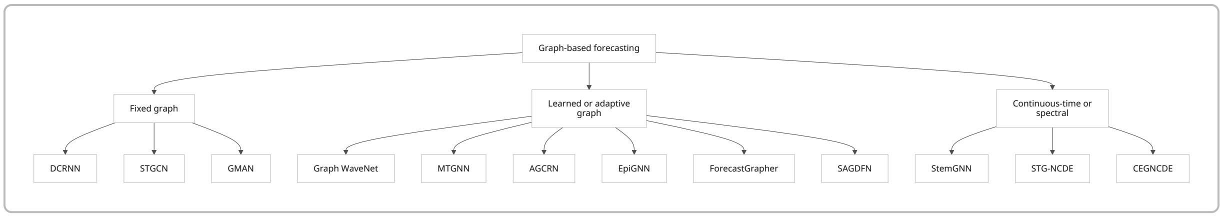
Across the literature, several architecture patterns recur. The most common recipe is a **spatial module plus a temporal module**: diffusion or spectral graph convolution for spatial dependence, and GRUs/LSTMs, temporal CNNs, attention, or Fourier operators for temporal dependence. A second recurring idea is that **the graph itself should be learnable**, because many forecasting settings have latent rather than known relationships; MTGNN, AGCRN, StemGNN, EpiGNN, CEGNCDE, and ForecastGrapher all stress this point in different ways. A third pattern is that long-horizon forecasting benefits from **non-autoregressive decoding or explicit horizon-to-history alignment**, such as transform attention in GMAN or direct multi-horizon outputs in Graph WaveNet and AGCRN. [2]

The main gaps are also remarkably consistent. Many papers still evaluate on a relatively narrow benchmark set dominated by traffic data; a large fraction rely on somewhat standardized splits and metrics rather than out-of-domain transfer, counterfactual shifts, or operational robustness tests. Interpretability of learned edges remains only partial: many models claim adaptive graph discovery, but only a minority give strong causal or semantic validation of the learned relations. Finally, results for newer preprints are often promising but less mature: some provide code and a clear conceptual advance, while others do not yet document enough benchmarking or reproducibility detail to displace the older, more established baselines. [3]

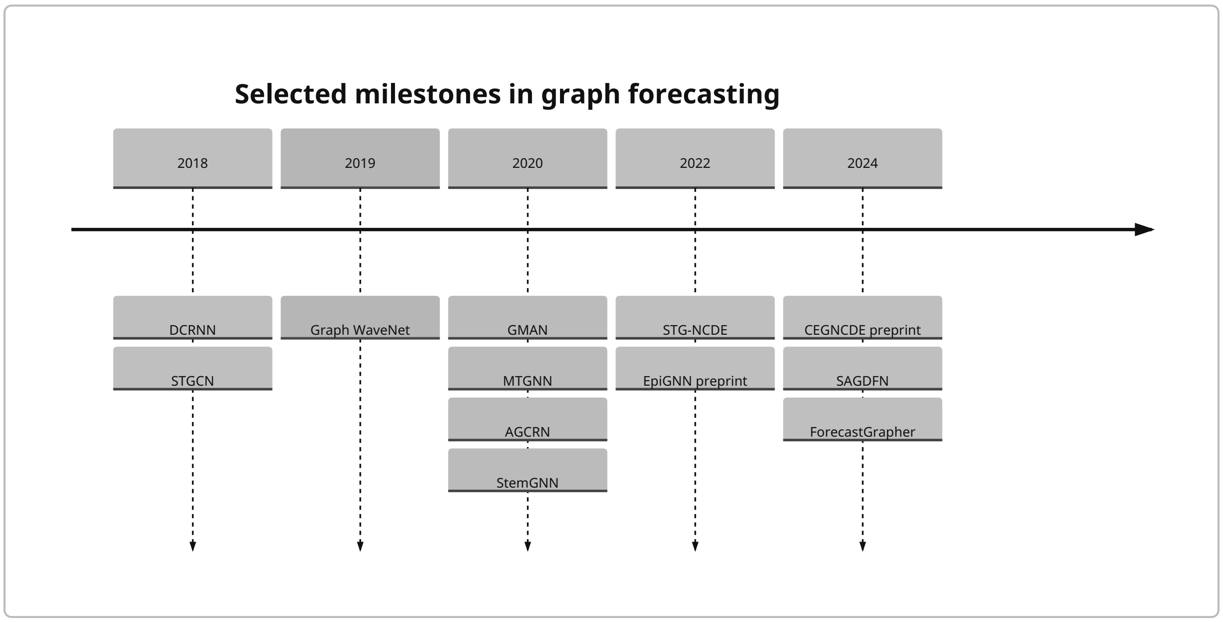
A practical reading list for someone entering the area is: **DCRNN → STGCN → Graph WaveNet → GMAN → MTGNN/AGCRN/StemGNN → STG-NCDE/EpiGNN → recent preprints such as CEGNCDE, SAGDFN, and ForecastGrapher**. That sequence mirrors the field's progression from fixed-graph spatio-temporal forecasting, to adaptive graph learning, to attention/spectral modeling, and then to continuous-time and scalability-focused methods. [4]

Taxonomy and reading map

The field can be organized into four broad families. **Graph plus recurrent models** include DCRNN, AGCRN, and STG-NCDE, where recurrent dynamics remain central but the message-passing operator is graphaware. **Graph plus temporal-convolution models** include STGCN, Graph WaveNet, MTGNN, and many scalability-oriented variants. **Attention-heavy models** include GMAN and epidemic graph learners such as EpiGNN. **Spectral or continuous-time models** include StemGNN and the NCDE family. [5]



A second useful lens is chronological. Early landmarks solved **traffic prediction on known road graphs**; around 2020, methods began to treat **general multivariate series as latent graphs**; after 2022, the literature moved toward **richer priors, irregular/continuous dynamics, and larger graphs**. [6]



Summary table

Paper	Year	Domain	Method	Main Contribution	Code?
DCRNN [7]	2018	Traffic	Diffusion convolution + seq2seq GRU + scheduled sampling	Put directed diffusion on road graphs at the center of spatio-temporal forecasting	Yes

Paper	Year	Domain	Method	Main Contribution	Code?
STGCN [8]	2018	Traffic	Spectral graph convolution + gated temporal CNN	Replaced recurrent temporal modules with a fully convolutional spatio-temporal block	Yes
Graph WaveNet [9]	2019	Traffic	Adaptive adjacency + dilated causal CNN + graph conv	Popularized self-adaptive graph learning with long receptive-field temporal convolutions	Yes
GMAN [10]	2020	Traffic	Spatio-temporal multi-head attention + transform attention	Used attention instead of graph convolution/ recurrence to tackle dynamic dependencies and long horizons	Yes
MTGNN [11]	2020	General MTS, traffic, electricity, solar	Graph learning + mix-hop propagation + dilated inception temporal conv	Framed general multivariate forecasting as learned directed graph forecasting without requiring a known topology	Yes
AGCRN [12]	2020	Traffic	Node-adaptive GCN + learned graph + recurrent decoder	Added node-specific parameter generation and adaptive graph construction	Yes

Paper	Year	Domain	Method	Main Contribution	Code?
StemGNN [13]	2020	Traffic, energy, ECG, COVID	Learned latent graph + GFT + DFT + spectral sequence cell	Jointly modeled inter-series and temporal dependence in the spectral domain	Yes
STG-NCDE [14]	2022	Traffic	Temporal NCDE + spatial NCDE/ graph processing	Brought continuous-time controlled differential equations to traffic graph forecasting	Yes
EpiGNN [15]	2022/2023	Epidemiology	Transmission-risk encoding + region-aware graph learner	Specialized graph learning for regional epidemic spread, including mobility-aware extensions	Yes
CEGNCDE [16]	2024	Traffic	Continuously evolving graph generator + GNCDE	Learned a graph that evolves continuously over time rather than staying static	Not located in sources reviewed
SAGDFN [17]	2024	Large-scale MTS	Scalable adaptive graph diffusion network	Targeted the scalability bottleneck of STGNNs on thousands of nodes	Not clearly linked in sources reviewed
ForecastGrapher [18]	2024	General MTS	Adaptive adjacency + node-regression framing + group feature convolution GNN	Reframed MTS forecasting explicitly as node regression to strengthen inter-series reasoning	Yes

Curated paper notes

Diffusion Convolutional Recurrent Neural Network

Citation and link. Yaguang Li, Rose Yu, Cyrus Shahabi, and Yan Liu. *Diffusion Convolutional Recurrent Neural Network: Data-Driven Traffic Forecasting.* ICLR 2018. Official paper and code are linked from OpenReview and the authors' GitHub. [7]

Main novelty. DCRNN made directed road-network diffusion the core spatial operator and paired it with a sequence-to-sequence recurrent forecaster using scheduled sampling, giving one of the first convincing graph-native long-horizon traffic forecasting systems. [19]

Methods. The model defines graph convolution through **bidirectional random walks** on a directed weighted graph, uses that operator inside a **diffusion convolutional GRU**, and trains a **seq2seq** forecaster with **scheduled sampling** to reduce exposure bias. The paper trains with neural objectives and reports MAE, RMSE, and MAPE for multi-step forecasting. [20]

Domain, datasets, code. Domain: traffic speed forecasting. Datasets: **METR-LA** and **PEMS-BAY**. The official TensorFlow implementation and data links are available in the authors' repository. [21]

Key quantitative results. On **METR-LA**, DCRNN reports **MAE/RMSE/MAPE = 2.77/5.38/7.3%** at 15 minutes, **3.15/6.45/8.8%** at 30 minutes, and **3.60/7.59/10.5%** at 1 hour, beating FC-LSTM and classical baselines. On **PEMS-BAY**, it reports **1.38/2.95/2.9%** at 15 minutes, **1.74/3.97/3.9%** at 30 minutes, and **2.07/4.74/4.9%** at 1 hour. The abstract summarizes this as a **12%–15% improvement** over prior state of the art. [22]

Limitations and open questions. DCRNN still assumes a largely fixed graph and can be computationally heavy because of recurrent decoding; later work mainly improved on exactly those two issues. [23]

Spatio-Temporal Graph Convolutional Networks

Citation and link. Bing Yu, Haoteng Yin, and Zhanxing Zhu. *Spatio-Temporal Graph Convolutional Networks: A Deep Learning Framework for Traffic Forecasting.* IJCAI 2018. An official IJCAI paper and open-source implementations are available. [8]

Main novelty. STGCN showed that one could replace recurrent temporal modeling with a **fully convolutional spatio-temporal block**, making training much faster while preserving good forecasting accuracy. [24]

Methods. STGCN interleaves **gated temporal 1D convolutions** with **spectral graph convolutions** inside an **ST-Conv block**. The paper explores Chebyshev and first-order graph approximations, trains using **mean squared error**, and evaluates with MAE, RMSE, and MAPE. [25]

Domain, datasets, code. Domain: traffic flow/speed forecasting. Datasets: **BJER4** and **PeMSD7** in medium and large settings. Public implementations exist, including the PyTorch STGCN repository. [26]

Key quantitative results. On **PeMSD7(M)**, **STGCN(Cheb)** reports **MAE/RMSE/MAPE = 2.25/4.04/5.26%** at 15 minutes and **3.57/6.77/8.69%** at 45 minutes, outperforming GCGRU and FC-LSTM. On **PeMSD7(L)**, it reports **2.37/4.32/5.56%** at 15 minutes and **3.97/7.45/9.73%** at 45 minutes. The paper also highlights major training-speed gains over GCGRU: about **272 s vs. 3825 s** on PeMSD7(M). [27]

Limitations and open questions. The model depends on a predefined topology and fixed spectral filtering, so later work targeted hidden dependencies, dynamic graphs, and longer-range temporal structure more explicitly. [28]

Graph WaveNet

Citation and link. Zonghan Wu, Shirui Pan, Guodong Long, Jing Jiang, and Chengqi Zhang. *Graph WaveNet for Deep Spatial-Temporal Graph Modeling.* IJCAI 2019. The IJCAI paper points to the public code repository. [9]

Main novelty. Graph WaveNet popularized the idea that a forecasting model should learn a **self-adaptive adjacency matrix** jointly with a temporal model that has a very large receptive field. [9]

Methods. The architecture combines **diffusion-style graph convolution** with a **learned adaptive dependency matrix** derived from node embeddings and **stacked dilated causal temporal convolutions** inspired by WaveNet. It trains with **MAE** and evaluates with MAE, RMSE, and MAPE. [9]

Domain, datasets, code. Domain: traffic speed forecasting. Datasets: **METR-LA** and **PEMS-BAY**. The original paper points to the public Graph WaveNet code. [9]

Key quantitative results. On **METR-LA**, Graph WaveNet reports **2.69/5.15/6.90%** at 15 minutes, **3.07/6.22/8.37%** at 30 minutes, and **3.53/7.37/10.01%** at 60 minutes, improving over DCRNN and STGCN. On **PEMS-BAY**, it reports **1.30/2.74/2.73%**, **1.63/3.70/3.67%**, and **1.95/4.52/4.63%** respectively. It also reduces inference cost substantially relative to DCRNN. [9]

Limitations and open questions. The adaptive graph is helpful, but it is still largely static once learned; subsequent work asked whether dependencies should vary with time or horizon. [29]

GMAN

Citation and link. Chuanpan Zheng, Xiaoliang Fan, Cheng Wang, and Jianzhong Qi. *GMAN: A Graph Multi-Attention Network for Traffic Prediction.* AAAI 2020. The AAAI paper links to the authors' GitHub repository. [10]

Main novelty. GMAN replaced most explicit graph-convolution/recurrent machinery with **spatio-temporal attention blocks** and introduced **transform attention** to map historical representations directly to future horizons, explicitly targeting long-horizon error propagation. [10]

Methods. The model uses an **encoder-decoder** with stacks of **spatial attention**, **temporal attention**, and **gated fusion** blocks, then inserts a **transform attention** layer between encoder and decoder. It trains by minimizing **MAE**. [30]

Domain, datasets, code. Domain: traffic volume and traffic speed. Datasets: **Xiamen** and **PeMS**. The paper states that source code is publicly available. [10]

Key quantitative results. On **Xiamen**, GMAN reports **MAE/RMSE/MAPE = 12.79/24.15/15.84%** at 1 hour, improving over Graph WaveNet's **13.33/24.77/16.50%**. On **PeMS**, GMAN reports **1.86/4.32/4.31%** at 1 hour, better than Graph WaveNet's **1.95/4.52/4.63%**. The abstract summarizes the 1-hour gain as **up to 4% MAE improvement** over prior work. [31]

Limitations and open questions. GMAN is elegant and effective, but it is still traffic-centric and depends on the inductive bias that attention alone can reliably recover spatial structure; generalization beyond traffic is less thoroughly documented in the original paper. [30]

MTGNN

Citation and link. Zonghan Wu, Shirui Pan, Guodong Long, Jing Jiang, Xiaojun Chang, and Chengqi Zhang. *Connecting the Dots: Multivariate Time Series Forecasting with Graph Neural Networks.* KDD 2020. A Monash publication page and the arXiv version provide the official paper metadata; the arXiv page links to the PDF and associated code resources. [32]

Main novelty. MTGNN was a major step in generalizing graph forecasting beyond traffic by learning a **directed latent graph among time series variables** even when no graph is provided. [33]

Methods. Core components are a **graph learning layer**, a **mix-hop propagation layer** designed to avoid oversmoothing while capturing multi-hop relations, and a **dilated inception temporal module** for multi-scale temporal patterns. It minimizes an **absolute loss with L2 regularization**. [34]

Domain, datasets, code. Domains: general multivariate time series and traffic. Single-step datasets include **Traffic**, **Solar-Energy**, **Electricity**, and **Exchange-Rate**; multi-step datasets include **METR-LA** and **PEMS-BAY**. The arXiv page provides code-discovery links. [35]

Key quantitative results. In single-step forecasting, MTGNN achieves strong results on **Traffic** with **RSE = 0.4162/0.4754/0.4461/0.4535** at horizons 3/6/12/24 and improves traffic RSE by **7.24%, 3.88%, and 4.83%** over the best baseline at horizons 3, 12, and 24, respectively. In multi-step traffic forecasting, it is competitive on **METR-LA** with **2.69/5.18/6.86%** at horizon 3 and **3.49/7.23/9.87%** at horizon 12, and on **PEMS-BAY** with **1.32/2.79/2.77%** at horizon 3 and **1.94/4.49/4.53%** at horizon 12. [36]

Limitations and open questions. The paper itself notes weaker gains on **Exchange-Rate**, hinting that graph learning helps most when there really is rich relational structure and enough data to learn it. [36]

AGCRN

Citation and link. Lei Bai, Lina Yao, Can Li, Xianzhi Wang, and Can Wang. *Adaptive Graph Convolutional Recurrent Network for Traffic Forecasting.* NeurIPS 2020. The official NeurIPS paper and repository are both available. [37]

Main novelty. AGCRN argued that beyond learning the graph, a good forecasting model should also learn **node-specific parameters**, not just shared global filters. [38]

Methods. The paper introduces **Node Adaptive Parameter Learning (NAPL)** to generate node-specific graph-convolution parameters from learned embeddings, plus **Data Adaptive Graph Generation (DAGG)** to learn the graph itself. Those modules are integrated into a recurrent forecasting architecture. [39]

Domain, datasets, code. Domain: traffic flow forecasting. Datasets: **PeMSD4** and **PeMSD8**. Official code and sample data are provided by the authors. [40]

Key quantitative results. Averaged over 12 horizons, AGCRN reports **MAE/RMSE/MAPE = 19.83/32.26/12.97%** on **PeMSD4**, beating STSGCN's **21.19/33.65/13.90%** and DCRNN's **21.22/33.44/14.17%**. On **PeMSD8**, it reports **15.95/25.22/10.09%**, improving over STSGCN's **17.13/26.86/10.96%** and DCRNN's **16.82/26.36/10.92%**. The paper summarizes this as **more than 5% relative improvements** in MAE and MAPE over the strongest baseline on both datasets. [41]

Limitations and open questions. Performance gains come with more parameters; the authors note that with embedding dimension 10, AGCRN has about **5×** the parameters of DCRNN, raising questions about scalability and overfitting on smaller datasets. [42]

StemGNN

Citation and link. Defu Cao, Yujing Wang, Juanyong Duan, Ce Zhang, Xia Zhu, Congrui Huang, Yunhai Tong, Bixiong Xu, Jing Bai, Jie Tong, and Qi Zhang. *Spectral Temporal Graph Neural Network for Multivariate Time-series Forecasting.* NeurIPS 2020. Official NeurIPS and Microsoft Research pages plus the official GitHub repository are available. [43]

Main novelty. StemGNN jointly models **inter-series structure and temporal evolution in the spectral domain**, combining **graph Fourier transform** and **discrete Fourier transform** in one end-to-end forecasting block. [44]

Methods. The model learns latent correlations, applies **GFT** to represent graph structure spectrally, applies **DFT** to capture temporal frequency patterns, then uses convolutional and sequence-style layers in the spectral domain. It includes **forecasting and backcasting branches** sharing an encoder. [44]

Domain, datasets, code. Domains: traffic, solar energy, electricity, ECG, and a COVID-19 case study. Datasets include **METR-LA, PEMS-BAY, PEMS07, PEMS03, PEMS04, PEMS08, Solar, Electricity, ECG5000, and COVID-19**. Microsoft released the official code. [45]

Key quantitative results. StemGNN reports **2.56/5.06/6.46%** on **METR-LA**, **1.23/2.48/2.63%** on **PEMS-BAY**, **2.14/4.01/5.01%** on **PEMS07**, **14.32/21.64/16.24%** on **PEMS03**, **20.24/32.15/10.03%** on **PEMS04**, **15.83/24.93/9.26%** on **PEMS08**, **0.03/0.07/11.55** on **Solar**, and **0.04/0.06/14.77** on **Electricity**. The paper summarizes the average gain as **8.1% on MAE** and **13.3% on RMSE** over the best baseline across datasets. For COVID-19, it reports **MAPE = 15.5/17.1/19.3** at 7/14/28-day horizons. [46]

Limitations and open questions. The spectral formulation is elegant, but its interpretability depends on the meaning of learned spectral components rather than raw edges, and the $O(N^3)$ eigendecomposition noted in the paper can become a bottleneck for very large graphs. [44]

STG-NCDE

Citation and link. Jeongwhan Choi, Hwangyong Choi, Jeehyun Hwang, and Noseong Park. *Graph Neural Controlled Differential Equations for Traffic Forecasting.* AAAI 2022. The official AAAI page and the authors' GitHub repository are available. [14]

Main novelty. STG-NCDE imported **neural controlled differential equations** into graph forecasting, modeling both temporal and spatial dynamics in continuous time. [47]

Methods. The model combines one NCDE for **temporal processing** and another for **spatial processing**, producing a continuous-time alternative to the usual graph-convolution-plus-RNN pattern. The official repository provides reproducibility scripts for multiple PeMS datasets. [14]

Domain, datasets, code. Domain: traffic forecasting. The paper states experiments on **6 benchmark datasets** with **20 baselines**, and the official repository includes commands for datasets such as **PEMSD4**. Official code is available. [14]

Key quantitative results. In the sources I reviewed, the most reliable public claim is the paper's own summary that STG-NCDE achieved the **best accuracy on all six datasets**, beating **20 baselines** by non-trivial margins. I was not able to recover the full metric tables from the official AAAI PDF within the available sources, so exact MAE/RMSE/MAPE numbers are incomplete here. [47]

Limitations and open questions. The central open question is whether the extra complexity of continuous-time modeling pays off consistently in operational settings versus strong discrete-time baselines such as Graph WaveNet or AGCRN. Reproducibility is helped by the public code, but the official public snippet I accessed did not surface the full result table cleanly. [48]

EpiGNN

Citation and link. Feng Xie, Zhong Zhang, Liang Li, Bin Zhou, and Yusong Tan. *EpiGNN: Exploring Spatial Transmission with Graph Neural Network for Regional Epidemic Forecasting.* Initially released as an arXiv preprint in 2022 and later published in ECML-PKDD/LNCS in 2023. The abstract also points to a public GitHub repository. [49]

Main novelty. EpiGNN specialized graph forecasting to epidemic spread by building a **region-aware graph learner** that is explicitly informed by **transmission risk**, geography, temporal signals, and optionally mobility data. [15]

Methods. The architecture combines a **transmission risk encoding module** with a **Region-Aware Graph Learner (RAGL)** and graph-neural forecasting components. The paper emphasizes the ability to fuse external spatial resources such as human mobility. [15]

Domain, datasets, code. Domain: epidemic forecasting. Datasets: **five real-world epidemic-related datasets**, including **influenza** and **COVID-19**. The abstract explicitly states that code and datasets are available in a public GitHub repository. [50]

Key quantitative results. The most robust result exposed in the sources I reviewed is the paper's headline claim that EpiGNN **outperforms state-of-the-art baselines by 9.48% in RMSE** across its epidemic benchmarks. [15]

Limitations and open questions. Epidemic forecasting is notoriously sensitive to reporting bias, policy shifts, and exogenous mobility interventions, so the long-term stability of learned spatial relations is an open question. EpiGNN is very compelling for public-health forecasting, but it is also more domain-shaped than general MTS methods like MTGNN or StemGNN. [51]

CEGNCDE

Citation and link. Jiajia Wu and Ling Chen. *Continuously Evolving Graph Neural Controlled Differential Equations for Traffic Forecasting.* arXiv 2024 preprint. [16]

Main novelty. CEGNCDE's key idea is that the **graph itself should evolve continuously over time**, not merely the node states. [16]

Methods. The paper introduces a **continuously evolving graph generator (CEGG)** built with NCDE ideas and combines it with a **graph neural controlled differential equation** forecaster to jointly model time continuity and changing spatial structure. [16]

Domain, datasets, code. Domain: traffic forecasting. The abstract reports extensive experiments, but the specific benchmark names and code availability were not surfaced in the sources I reviewed. [16]

Key quantitative results. The abstract reports average gains of **2.34% relative MAE reduction**, **0.97% relative RMSE reduction**, and **3.17% relative MAPE reduction** over prior state of the art. [16]

Limitations and open questions. Because this is still a preprint in the sources I reviewed, the main open questions are reproducibility, code availability, and whether the improvements remain robust across standard traffic leaderboards. [16]

SAGDFN

Citation and link. Yue Jiang, Xiucheng Li, Yile Chen, Shuai Liu, Weilong Kong, Antonis F. Lentzakis, and Gao Cong. *SAGDFN: A Scalable Adaptive Graph Diffusion Forecasting Network for Multivariate Time Series Forecasting.* arXiv 2024; the ST-Dataminer lab reports it as accepted at **ICDE 2024**. [17]

Main novelty. SAGDFN squarely targets **scalability** for graph-based multivariate forecasting, especially on datasets with **thousands of nodes**, where many STGNNs become memory- or compute-bound. [52]

Methods. It uses a **scalable adaptive graph diffusion** design intended to preserve complex spatial-temporal correlation even when no prior graph is given. The paper positions itself as a graph-diffusion system for large-scale MTS rather than primarily a traffic-specific model. [52]

Domain, datasets, code. Domain: large-scale multivariate time series forecasting. The abstract says the method is evaluated on **one 207-node real-world dataset** and **three 2000-node datasets**. In the sources I reviewed, I did not find an official code repository clearly linked. [17]

Key quantitative results. The strongest public claim available in the accessed sources is that SAGDFN is **comparable to prior SOTA on the 207-node dataset** and **outperforms all SOTA baselines by a significant margin on three 2000-node datasets**. Exact dataset names and metric values were not fully surfaced in the source excerpts I reviewed. [52]

Limitations and open questions. The key question is whether the scalability gains come with any trade-off in interpretability or small-data performance. Benchmark transparency is also still less mature than for older flagship papers. [52]

ForecastGrapher

Citation and link. Wanlin Cai, Kun Wang, Hao Wu, Xiaoxu Chen, and Yuankai Wu. *ForecastGrapher: Redefining Multivariate Time Series Forecasting with Graph Neural Networks.* arXiv 2024, with a public GitHub repository. [18]

Main novelty. ForecastGrapher makes a conceptual move: it reframes multivariate time series forecasting explicitly as a **node regression** problem and augments GNN expressivity with a **group feature convolution** mechanism. [53]

Methods. The pipeline includes custom node embeddings, **adaptive adjacency learning**, and a proposed **Group Feature Convolution GNN (GFC-GNN)** that splits features into groups and applies different 1D convolutions before aggregation. [53]

Domain, datasets, code. Domain: general multivariate time series forecasting. Official code is available on GitHub. The abstract does not enumerate datasets in the source excerpt I reviewed, though the repository exposes the implementation structure for running experiments. [54]

Key quantitative results. The strongest high-confidence claim available from the sources I reviewed is qualitative: ForecastGrapher **surpasses strong baselines and leading published methods** in MTS forecasting. I was not able to recover the metric table from the official paper excerpt within the available sources, so the exact numbers are incomplete here. [53]

Limitations and open questions. As with several recent preprints, the main limitation is that the abstract is clearer on the idea than on benchmark specifics in the sources I reviewed. The framing is intellectually interesting, but the degree to which it changes practice versus improving implementation details remains to be validated over time. [18]

Cross-paper trends and research gaps

A robust empirical pattern is that **learned or adaptive graphs usually beat fixed graphs** when the true dependency structure is only partially known. Graph WaveNet's adaptive adjacency, AGCRN's DAGG, MTGNN's learned directed graph, StemGNN's latent correlation layer, EpiGNN's region-aware learner, and CEGNCDE's continuously evolving graph generator are all variants of that same claim. The field's consensus, at this point, is not that graphs are helpful only when known a priori, but that **graph learning itself is part of forecasting**. [55]

Another clear trend is that **temporal modeling moved from recurrence toward richer alternatives**. DCRNN and AGCRN are still recurrent; STGCN and Graph WaveNet use temporal convolutions; GMAN uses attention; StemGNN moves to frequency-domain temporal operators; STG-NCDE and CEGNCDE move to continuous-time trajectories. This is one of the clearest macro-shifts in the literature. [56]

The most important gap is **benchmark concentration**. Traffic remains the dominant test bed, which is understandable because road sensors form obvious graphs, but it means that claims of broad superiority are sometimes stronger than the real evidence. General MTS papers like MTGNN and StemGNN help, and domain-specific papers like EpiGNN show the paradigm can transfer, but there is still relatively less canonical work in retail, energy-system operations beyond standard electricity benchmarks, and finance with transparent public benchmarks. [57]

A second gap is **interpretability**. Many models learn edges, but few convincingly establish whether those edges correspond to actionable interaction structure, confounding, or just useful predictive shortcuts. AGCRN, StemGNN, and EpiGNN take interpretability more seriously than many peers, but the broader literature still leans toward predictive usefulness over semantic validation. [58]

A third gap is **reproducibility for newer preprints**. Older landmarks such as DCRNN, STGCN, Graph WaveNet, AGCRN, StemGNN, and STG-NCDE all have visible public code in the sources reviewed. For some newer papers, especially the 2024 preprints, code or full benchmark tables were not clearly surfaced in the accessible official sources I reviewed, which makes them interesting but somewhat harder to treat as canonical without caution. [59]

Open questions and caveats

Some entries above have **incomplete quantitative detail** because the official public sources I reviewed did not surface the full result tables cleanly enough within the available excerpts, especially for **STG-NCDE**, **SAGDFN**, and **ForecastGrapher**. Where that happened, I explicitly reported only the claims I could support from the primary paper pages, abstracts, or official repositories. [60]

If you want the single best "core syllabus," the highest-confidence papers to prioritize are **DCRNN, STGCN, Graph WaveNet, GMAN, MTGNN, AGCRN, and StemGNN**. Those seven cover the most important architectural ideas in the literature and have the strongest combination of influence, benchmarking, and code availability in the sources reviewed. [61]

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1 4 6 7 19 61 Diffusion Convolutional Recurrent Neural Network: Data-Driven Traffic Forecasting | OpenReview
https://openreview.net/forum?id=SJiHXGWAZ&utm_source=chatgpt.com

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